VesperGrid

Critical Infrastructure Operational Twin on AMD MI300X

A source-linked decision-support model for industrial safety incidents.
Multimodal evidence in. Candidate plans, with audit trail, out.

`Track 3 · Qwen Challenge · HF Special Prize · Build-in-Public`

The 18-minute problem

A port operator gets four signals at once:

→ one **drone keyframe**

→ one **gate-camera clip**

→ one **wind-sensor strip**

→ one **operator radio note**

They contradict each other. There is no single source of truth.
Decision pressure is rising. A wrong route can cost lives.

What VesperGrid does

> “Turn **fragmented multimodal evidence** into a **source-linked operational twin**, with uncertainty named, not buried.”

```
evidence pack → async ingest → Qwen-VL on MI300X
            → normalized graph → candidate plan + uncertainty ledger
            → every claim links back to a SRC-* UUID
```

The console is the operator's read-only view of the evidence-to-decision chain.

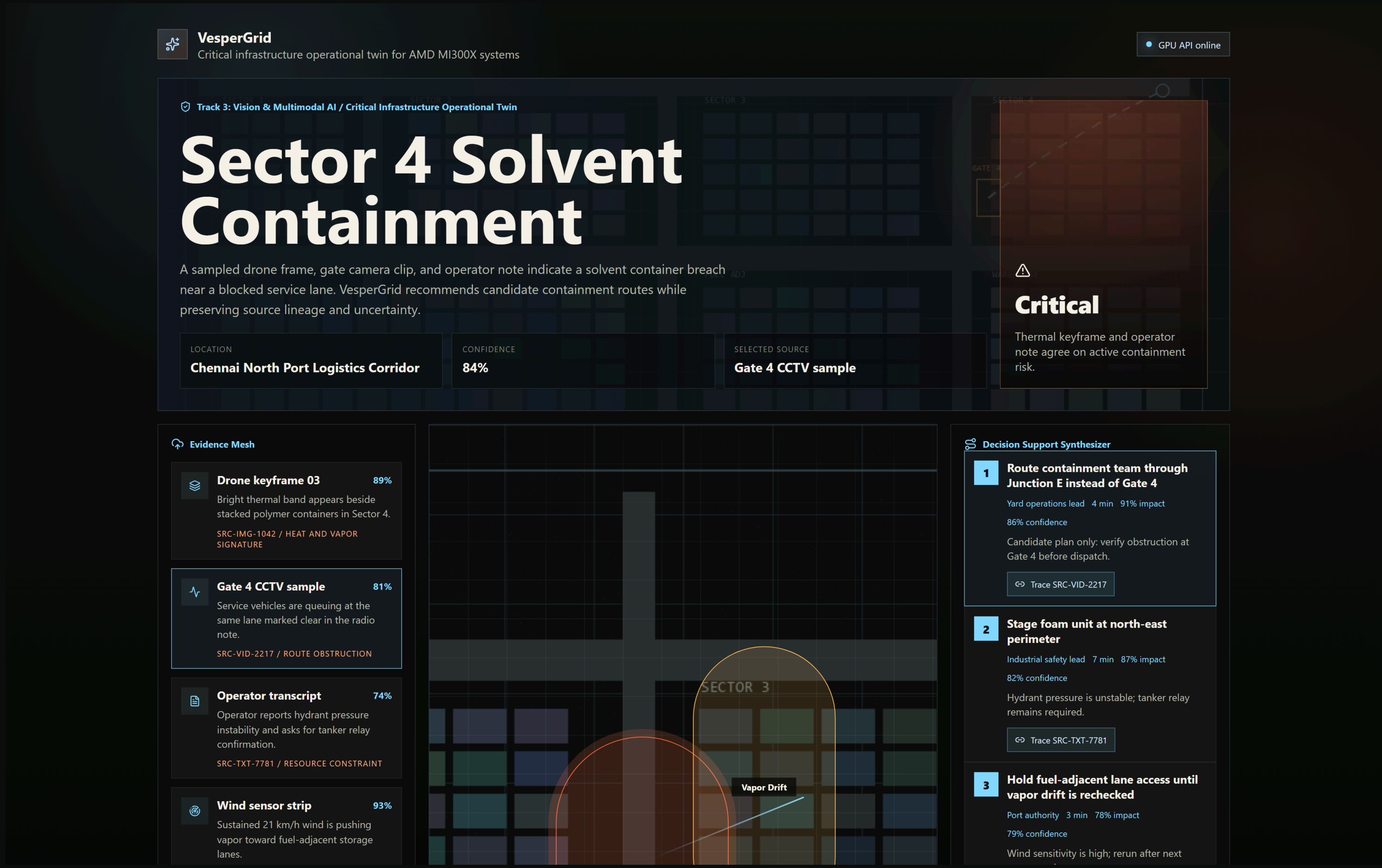
Operations console

→ **Hero band** — incident, severity, confidence

→ **Evidence Mesh** — four sources, each `SRC-*` tagged

→ **Decision Synthesizer** — plans linked to evidence

→ Live ingest appends a fifth source: the operator note



Source lineage — the wow moment

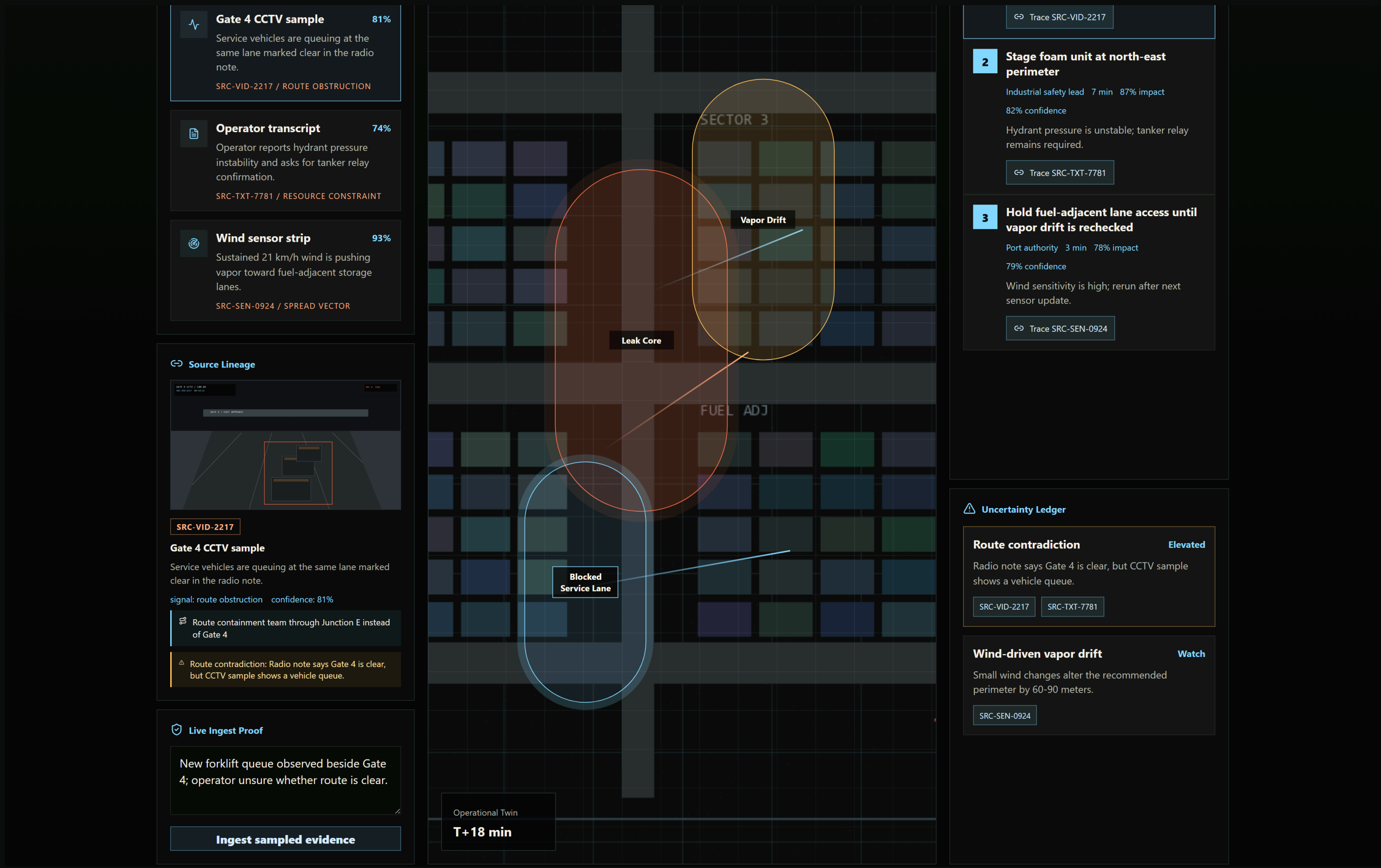
Click `SRC-VID-2217`. One UUID surfaces three things:

→ the **raw CCTV thumbnail** (what the AI saw)

→ the **candidate plan** it produced

→ the **uncertainty issue** it created

No prose. No prompt magic. A verifiable chain.



Async pipeline — no synchronous trap

```
POST /api/ingest           → { job_id }
GET  /api/ingest/{id}/events  (SSE)

queued → sampling → parsing → normalizing → synthesizing → complete
```

Per-stage SSE events drive a real progress bar in the console.

If the GPU backend fails mid-pipeline, the API soft-degrades to a deterministic synthesizer. The demo never breaks. Schemas are Pydantic-validated end-to-end.

Why AMD MI300X

Resource	Value
GPU	1× AMD Instinct MI300X
GPU memory	**192 GB VRAM**
Model	Qwen2.5-VL-7B-Instruct via vLLM ROCm
Tensor parallel	TP=1 (single GPU, zero NCCL/RCCL surface)
Multi-image budget	5 keyframes per prompt, resident

192 GB on one card means a frontier multimodal model stays warm with full multi-image context between operator turns. **Latency, not FLOPs, is the bottleneck for an operational twin.**

Honesty

What is **GPU-resident** on the MI300X:

→ Qwen-VL multimodal inference (paged attention, KV cache, multi-image decode)

What is intentionally **CPU-vectorized** for the MVP:

→ bounded route/hazard scoring

→ in-memory entity graph

The Uncertainty Ledger is **first-class**, not an afterthought. Every contradiction is named, with a severity label and a link back to the offending source.

Business value

Buyers feel the pain in cost-per-minute of downtime:

→ **Ports & logistics yards** — vessel queuing, gate dwell, dangerous-goods routing

→ **Energy & petrochemical** — perimeter incidents, vapor drift, evacuation paths

→ **Heavy industry** — forklift collisions, contained-spill protocols

Roadmap (post-hackathon)

→ real evidence-graph backing store (Neo4j / DuckDB-vector)

→ per-tenant policy packs + role-based access

→ frontier-model fine-tunes on synthetic operator transcripts

Submission map

Asset	Location
Code repository	`<github-url>`
Live app (HF Space)	`<huggingface.co/spaces/...>`
AMD Cloud deploy	`scripts/bootstrap_amd_cloud.sh` (idempotent, single MI300X)
Demo evidence pack	`demo/sector4/` (synthetic, license-clean)
Live tracker	`PROGRESS.md`

> **Stackable prizes claimed:** Qwen Challenge · Hugging Face Special Prize · Ship It / Build-in-Public

Thank you. Operators deserve evidence-grounded tools, not chatbots.